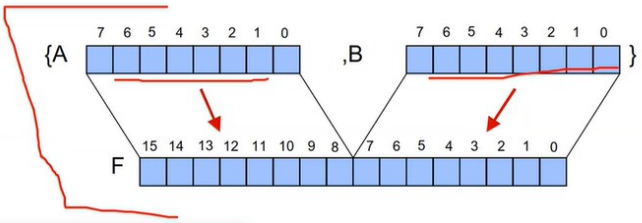
Nets: reg & wire

- physical wires(nets) are represented in Verilog using two datatypes
  - reg
  - wire

```
wire a, b, z;                  // three 1-bit wires
wire [31:0] memdata;           // a 34-bit bus
reg [7:0] b1, b2, b3, b4;      // four 8-bit buses
```

- Note that [7:0] means that 7 is the most significant bit & 0 is the least significant bit
- Concatenation using curly brackets can be used

```
wire [7:0] A, B;
reg [15:0] F;
F = {A, B}
```



37:14

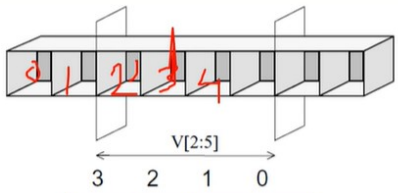
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Part select

```
reg [0:7] V;
reg [3:0] W;
```

```
W = V[2:5];
```

```
W[1:0] = 2'b11;
```



	3	2	1	0
W:	V[2]	V[3]	V[4]	V[5]

	3	2	1	0
W:	V[2]	V[3]	1	1

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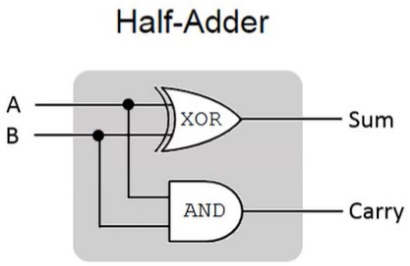
Procedural Block: always

- Two type of codes in almost all HDLs
  - Dataflow
  - Behavioral/Procedural
- Behavioral/Procedural
  - Statements are executed inside procedural blocks
  - Two of the most used procedural block are *always* and *initial* (later) blocks
  - Procedural blocks are executed in parallel
  - These blocks support C-like control structure such as **if, for, while, case**
  - Statements are executed in order like C code
  - **reg** data type **must** be declared for net on the left hand side of any assignment inside the procedural blocks
- always block
  - The statements inside the always block are executed sequentially
  - The statements are only executed whenever the signals inside the sensitivity list are updated
  - always @(*) can be used to trigger the block whenever any RHS nets are updated
  - Note that always @(a, b, c) is the same as always@(*)

```
always @ (<sensitivity_list>) begin
  .
  .
  <statements executed in order>
  .
  .
end
```


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Example: Half Adder




```
module half_adder(A, B, Sum, Carry);
  input A, B;
  output reg Sum, Carry;

  always @(*) begin

    Sum = A ^ B;
    Carry = A & B;

  end

endmodule
```


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If else

- If .. else if .. else statements execute a statement, or a block of statements depend on the result of the expression

Syntax

```
if (expression) begin
... statements ...
end
else if (expression) begin
... statements ...
end
...more else if blocks ...
else begin
... statements ...
end
```

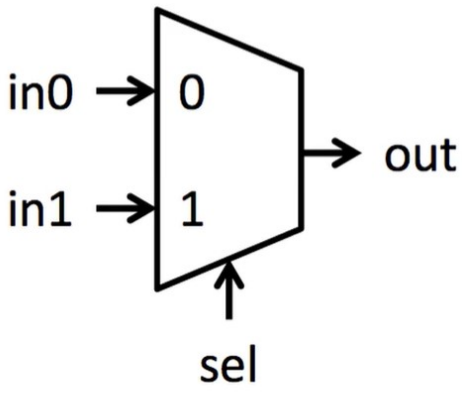
```
If (a == b) begin
  y = 2'b01;
  out = 4'b1000;
end
else begin
  y = 2'b00;
  out = 4'b00;
end
```

```
if (alu_op == 2'b00)
  aluout = a + b;
else if (alu_op == 2'b01)
  aluout = a - b;
else if (alu_op == 2'b10)
  aluout = a & b;
else // alu_op == 2'b11
  aluout = a | b;
```


45:54

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Example: 2:1 MUX




```
module mux2(in0, in1, sel, out);
  input in0, in1, sel;
  output reg out;

  always @(*) begin

    if (sel == 1) begin
      out = in1;
    end
    else if (sel == 0) begin
       out = in0;
    end

  end

endmodule
```


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Case

- Case allow multiple branch based on comparing the expression
- Statements in the default block is executed when none of the other case choices are true
- If the default are don't care define them as 'x' or '0'

```
case ({alu_op_0, alu_op_1})
  2'b00: aluout = a + b; // if alu_op_0, alu_op_1 is 2'b00
  2'b01: aluout = a - b;
  2'b10: aluout = a & b;
  default: aluout = 1'b0;
endcase
```

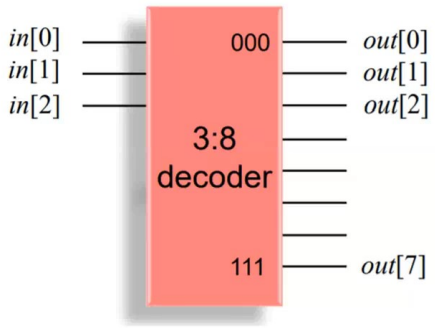
Syntax

case (expression)
case_choice1:
begin
... statements ...
end
case_choice2:
begin
... statements ...
end
... more case choices blocks ...
default:
begin
... statements ...
end
endcase

51:48

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Example: 3:8 Decoder



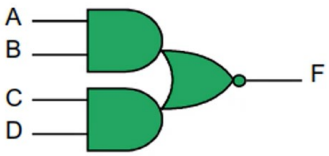

```
module decoder(in, out);
  input [2:0] in;
  output reg [7:0] out;

  always @(in) begin
    case(in)
      3'b000: out=8'b0000_0001;
      3'b001: out=8'b0000_0010;
      3'b010: out=8'b0000_0100;
      3'b011: out=8'b0000_1000;
      3'b100: out=8'b0001_0000;
      3'b101: out=8'b0010_0000;
      3'b110: out=8'b0100_0000;
      3'b111: out=8'b1000_0000;
    endcase
  end
```


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Continuous assignment




```
// An and-or-invert gate
module AOI (A, B, C, D, F);
  input A, B, C, D;
  output F;

  assign F = ~((A & B) | (C & D));
endmodule
```

&	and
|	or
~	not

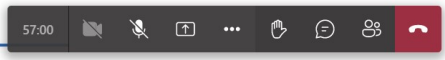
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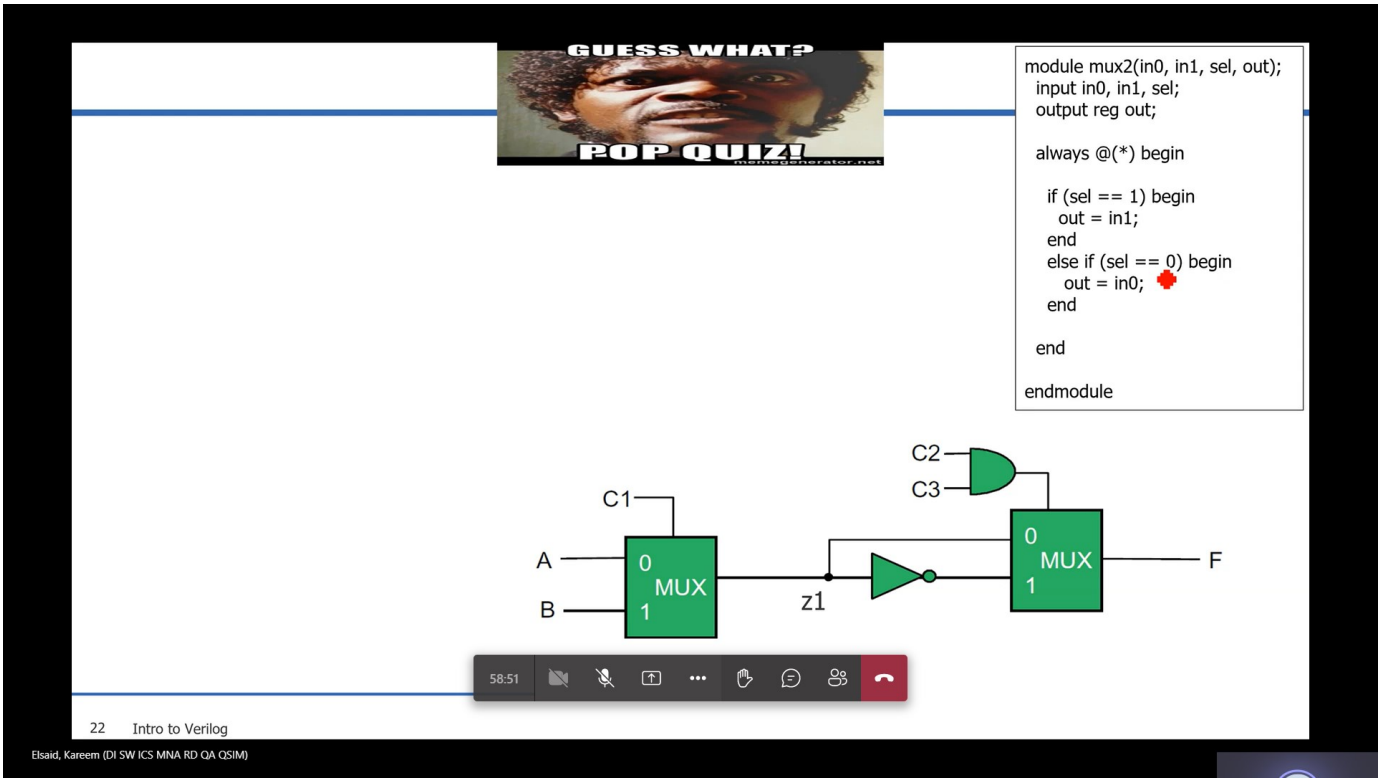
Mux using conditional operator

```
assign out = (sel == 1)? in1 : in0;
```

/* if the sel signal == 1 then "out = in1", else "out = in0" */



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Module Instantiations

Press Esc to exit full screen

- Module instances provide the ability to reuse any given module inside other modules
- Bigger and more complex designs are built by integrating multiple modules in a hierarchical manner
- Modules can be instantiated within other modules and ports of these instances can be connected with other signals inside the parent module
- For example, 2x1 mux can be implemented and then used multiple times inside a bigger designs instead of rewriting its functionality again
  - 4x1 MUX can be implemented using 2x1 MUX

```
<module_name>  <instance_name_1> (port_connection_list);
   mux2            m1          (.sel(sel[0]), .in0(d0), .in1(d1), .out(z1));   // MODULE PORT MAPPING BY NAME
OR
   mux2            m1          (sel[0], d0, d1, z1);          // MODULE PORT MAPPING BY ORDER
```

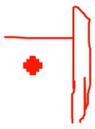
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Parameterized Modules

- Suppose we have need to have a MUX with variable bit width for the data to be selected
- Parameters provides the flexibility to change word length later in any module instantiation by changing only the parameter value
  - Parameter must be declared before the module ports



Syntax for declaring parameters of a module

parameter par_1 = *value*,
par_2 = *value*, ...;
parameter [range] parm_3 = value

Syntax to override parameters when instantiating a module

module_name #(1st_parameter_values,
2nd_parm_value, ...)
instance_name(port_connection_list);

```
module mux2(in0, in1, sel, out);
 parameter W2 = 1; // width of data is by default 1 bit
 input [W2-1:0] in0, in1;
 input sel;
 output [W2-1:0] out;

 assign out = (sel == 1)? in1 : in0;

endmodule
```

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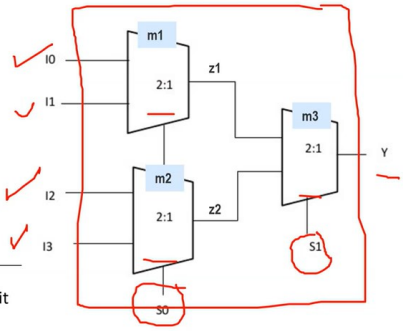
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Example: 4:1 MUX using 2:1 MUX

```
module mux2(in0, in1, sel, out);
  parameter W2 = 1; // width of data is by default 1 bit
  input [W2-1:0] in0, in1;
  input sel;
  output [W2-1:0] out;

  assign out = (sel == 1)? in1 : in0;

endmodule
```




```
module mux4(i0, i1, i2, i3, s, y);
parameter W4 = 1; // width of data is by default 1 bit
  input [W4-1:0] i0, i1, i2, i3;
  input [1:0] s;
  output [W4-1:0] y;

  wire z1, z2; // since z1, z2 aren't used inside procedural blocks then these nets are declared as wire

  mux2 #(.W2(W4)) m1(.sel(s[0]),.in0(i0),.in1(i1),.out(z1));
  mux2 #(.W2(W4)) m2(.sel(s[0]),.in0(i2),.in1(i3),.out(z2));
  mux2 #(.W2(W4)) m3(.sel(s[1]),.in0(z1),.in1(z2),.out(y));

endmodule
```

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Coding Guidelines

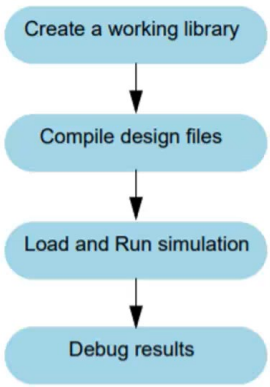
Rule #	Description	Example
1	If the name consists of multiple words, the words must be separated with underscores.	word_width
2	When possible, use the same name or similar names for ports and signals that are connected.	mem_out = mem_buf
3	Variables must use descriptive names	Incorrect: ra Correct: ram_addr
4	Must not use i, j, k, etc for variables other than loop counters.	
5	Code blocks must follow the following rules: The first delimiter 'begin' must be on the same line as of the enclosing statement Code within the code block must be indented	always@(*) begin ... End
6	Each block of code is indented one degree more than the enclosed block	always@(*) begin if(sel) .. end

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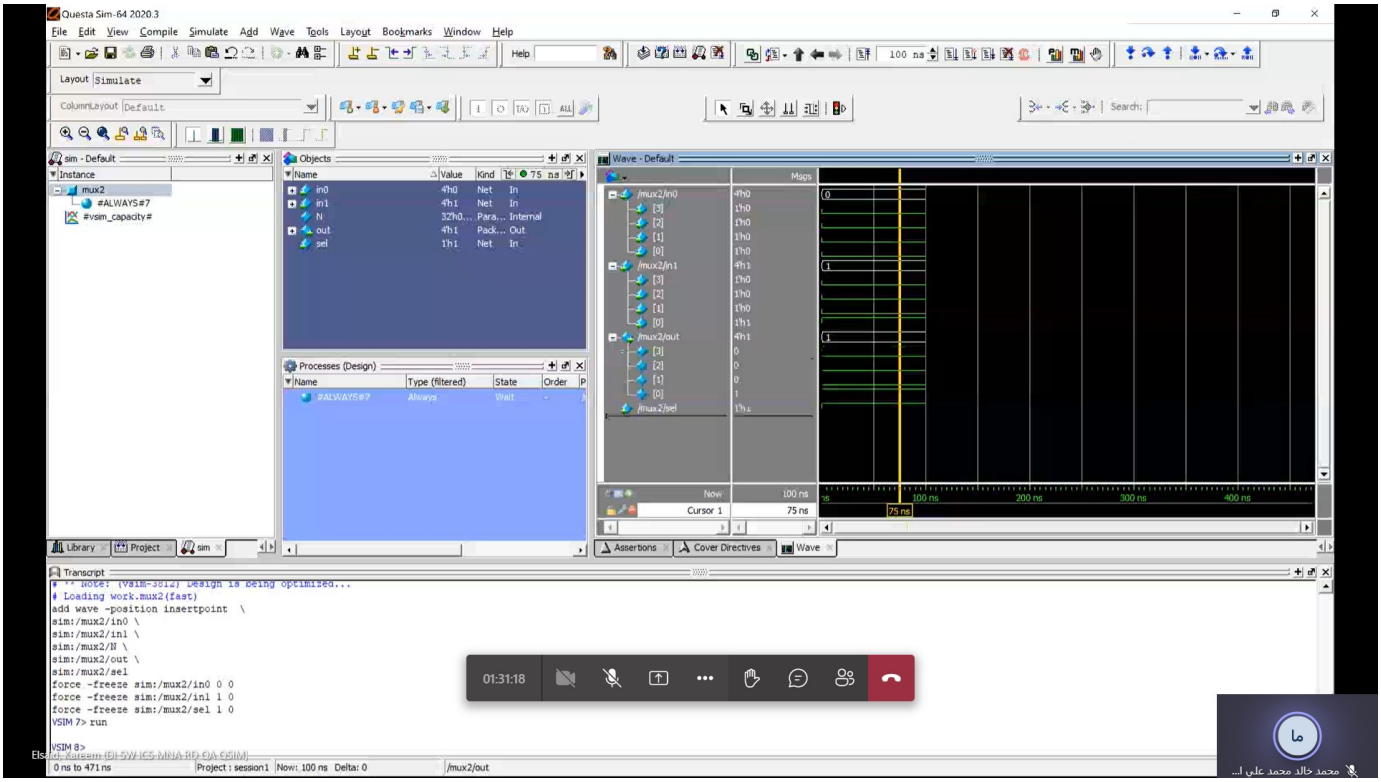
Basic simulation flow

- In ModelSim, all designs are compiled into a library. You typically start a new simulation in ModelSim by creating a working library called "work," which is the default library name used by the compiler as the default destination for compiled design units
- After creating the working library, you compile your design units into it.
- With the design compiled, you load the simulator with your design by invoking the simulator on a top-level module (typically the testbench)



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